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Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 14	
TITLE: Exception Handling using Try-Catch	

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

1. **Create an ATM program handling invalid withdrawal amount using try-catch**

Programme

```
ATM.java 3
C: > Users > RIKISHA > PROGRAMMING > Java > src > ATM.java > ATM
1   import java.util.Scanner;
2
3   public class ATM {
        Run | Debug
4       public static void main(String[] args) {
5           Scanner sc = new Scanner(System.in);
6
7           double balance = 10000;
8
9           try {
10              System.out.print(s: "Enter withdrawal amount: ");
11              double amount = sc.nextDouble();
12
13              if (amount <= 0) {
14                  throw new Exception(message: "Invalid withdrawal amount!");
15              }
16
17              if (amount > balance) {
18                  throw new Exception(message: "Insufficient balance!");
19              }
20
21              balance = balance - amount;
22
23              System.out.println(x: "Withdrawal successful.");
24              System.out.println("Amount withdrawn: " + amount);
25              System.out.println("Remaining balance: " + balance);
26          }
27          catch (Exception e) {
28              System.out.println("Error: " + e.getMessage());
29          }
30
31          sc.close();
32      }
33  }
```

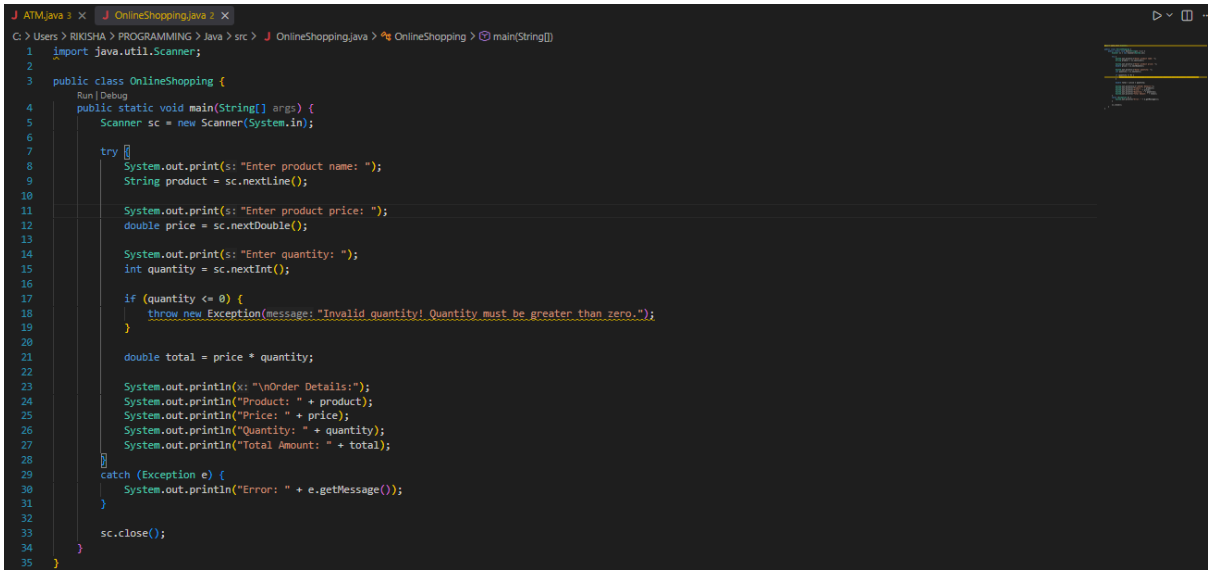
Output

```
Problems 3   Output   Debug Console   Terminal   Ports                                   powershell

PS C:\Users\RIKISHA> javac ATM.java
error: file not found: ATM.java
Usage: javac <options> <source files>
use --help for a list of possible options
PS C:\Users\RIKISHA> cd "C:\Users\RIKISHA\PROGRAMMING\Java\src"
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac ATM.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java ATM
Enter withdrawal amount: 250000
Error: Insufficient balance!
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

2. **Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.**

Program



```java
import java.util.Scanner;

public class OnlineShopping {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter product name: ");
            String product = sc.nextLine();

            System.out.print("Enter product price: ");
            double price = sc.nextDouble();

            System.out.print("Enter quantity: ");
            int quantity = sc.nextInt();

            if (quantity <= 0) {
                throw new Exception("Invalid quantity! Quantity must be greater than zero.");
            }

            double total = price * quantity;

            System.out.println("\nOrder Details:");
            System.out.println("Product: " + product);
            System.out.println("Price: " + price);
            System.out.println("Quantity: " + quantity);
            System.out.println("Total Amount: " + total);
        }
        catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output



```
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> javac OnlineShopping.java
PS C:\Users\RIKISHA\PROGRAMMING\Java\src> java OnlineShopping
Enter product name: Laptop
Enter product price: 70000000
Enter quantity: 2

Order Details:
Product: Laptop
Price: 7.0E7
Quantity: 2
Total Amount: 1.4E8
PS C:\Users\RIKISHA\PROGRAMMING\Java\src>
```

Conclusion

Thus, the programs successfully demonstrate exception handling in Java using try-catch. Invalid ATM withdrawal amounts and incorrect product quantities are handled appropriately, preventing program errors and displaying meaningful error messages to the user.